

Lecture 21: Continuous-Time Markov Chains

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Table of contents

- 1 Continuous-time Markov chains::: {.callout-note icon=false}## Definition: CTMCThe process stays in state i for an $\text{Exponential}(q_i)$ holding time (memoryless), then jumps to $j \neq i$ w.p. q_{ij}/q_i , where $q_i = \sum_{j \neq i} q_{ij}$. The rate matrix Q plays the role P played in discrete time.:... . ::: {.callout-warning icon=false}## FactLong-run behavior is again governed by a stationary distribution, satisfying $\pi Q = 0$:::—## Simulating a CTMC via the next-event method 1

—title: “Lecture 21: Continuous-Time Markov Chains”author: “Jamie Haddock”format: revealjs: output-file: Lecture21_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

- 1 Continuous-time Markov chains::: {.callout-note icon=false}## Definition: CTMCThe process stays in state i for an $\text{Exponential}(q_i)$ holding time (memoryless), then jumps to $j \neq i$ w.p. q_{ij}/q_i , where $q_i = \sum_{j \neq i} q_{ij}$. The rate matrix Q plays the role P played in discrete time.:... . ::: {.callout-warning icon=false}## FactLong-run behavior is again governed by a stationary distribution, satisfying $\pi Q = 0$:::—## Simulating a CTMC via the next-event method

```
q01, q10 = 0.5, 1.0T = 200.0t, state = 0.0, 0times, states = [0.0], [0]while t < T:    rate = q01 if st
durations = np.diff(times)time_in_state0 = durations[states[:-1] == 0].sum()print("empirical fraction o
```

We now shift from *modeling* randomness to *estimating* things from it.## Reading & HomeworkReading: §7.5 / supplementary notes (Bertsekas & Tsitsiklis, *Introduction to Probability*).